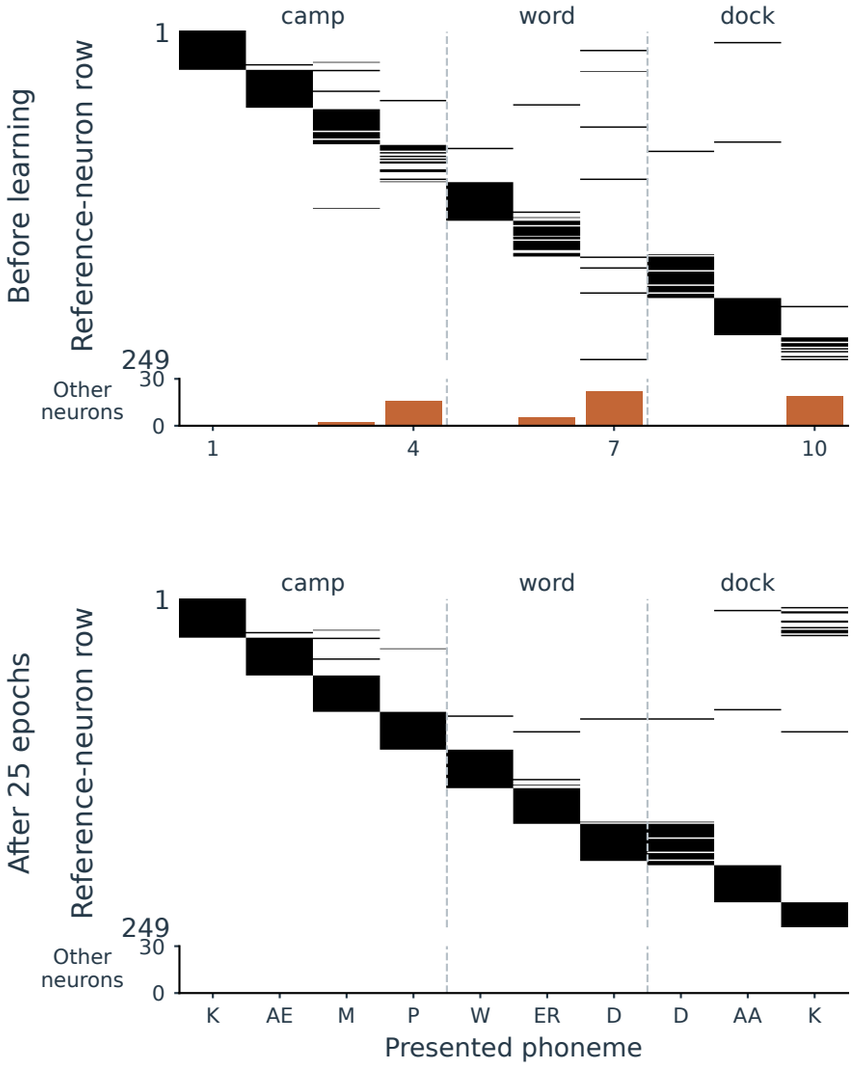
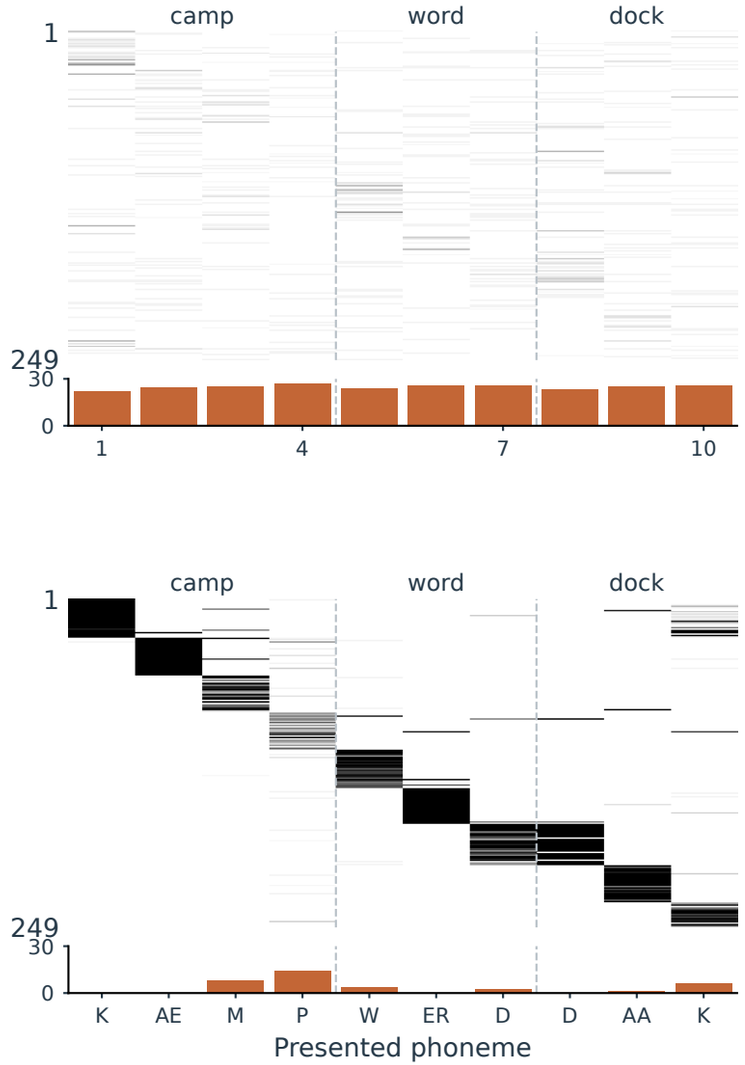
Learning stabilises neuronal sequences under noise

Three words, separately reset · fixed neuron order · same corrupted inputs before and after learning

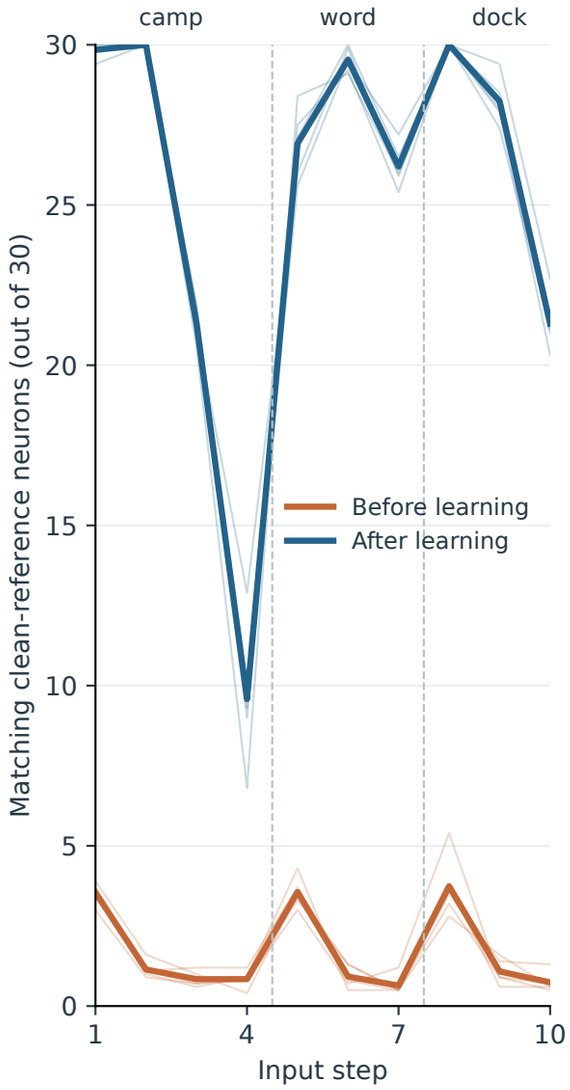
Clean replay



Corrupted replays



Across five seeds



Same three words · 10 probes per prefix
Thin lines = seed means

Never active — Always active

Raster: 249 neurons in the trained clean reference. Orange strips count all other neurons.

Noise replaces 6/30 input neurons at every phoneme. Shading: mean of 10 independent noisy prefix probes; no smoothing.

The clean band is created by ordering; its persistence in independent corrupted inputs—not its shape—is the evidence.